

LAB 1: REQUIREMENTS ENGINEERING & UML USE-CASE MODELLING

EXCEPTION FLOW SPECIFICATION

Problem Statement #24 — Smart Cities, Transport & Logistics

Public Bus Live Tracking & Crowding Estimator

NAME: SHWETHA S

SRN: PES1UG24CS452

SECTION: 'H'

Use Case: UC-03 — View Bus Information

Primary Actor: Commuter

Exception: Ticketing Sensor Data Failure

Goal:

Handle a failure or invalid response from the ticketing sensor system while allowing the commuter to continue viewing other available bus information.

Exception Flow E1 — Ticketing Sensor Failure

Trigger

The ticketing sensor system fails to provide valid passenger data for the selected bus.

Flow

1. The commuter selects a bus route or a specific bus.
2. The system retrieves the latest GPS information.
3. The system determines the current bus location.
4. The system calculates the estimated arrival time for upcoming stops.

5. The system attempts to retrieve passenger information from the ticketing sensors.
6. The system detects that the ticketing sensor data is unavailable or invalid.
7. The system cannot calculate a reliable current crowding level.
8. The system displays the following message:

"Crowding information is temporarily unavailable."

9. The system continues displaying the available bus location and ETA information.
10. The system continues monitoring the ticketing sensor system.
11. When valid ticketing sensor data becomes available, the system retrieves the new data.
12. The system calculates the updated crowding level.
13. The system displays the updated crowding information to the commuter.
14. The use case continues normally.

Exception Conditions

The following exception conditions may occur during the use case:

E1. Ticketing Sensor Failure

The ticketing sensor does not provide valid passenger information.

System Response:

Display a crowding-information-unavailable message and continue showing available location and ETA information.

E2. Invalid GPS Data

The received GPS coordinates are invalid or unusable.

System Response:

Do not display an unreliable current location. Inform the commuter that live location and ETA information are temporarily unavailable.

E3. Communication Failure

The system cannot communicate with an external GPS or ticketing sensor source.

System Response:

Display the latest available information where possible and continue attempting to retrieve new data.

Exception Flow Outcome

The system should fail gracefully when sensor information is unavailable. Instead of terminating the entire use case, it should provide the information that remains available and clearly inform the commuter about the unavailable information.